

Areeb Ehsan

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science, Siebel School of Computing and Data Science

Urbana-Champaign, IL

Aug 2026 – May 2028

Georgia State University

Atlanta, GA

B.S. Computer Science, Data Science Certificate | **GPA: 3.96** | **Summa Cum Laude** | **Distinction in Major**

May 2026

RESEARCH & PUBLICATION

AI-Enhanced Virtual Biopsies for Brain Tumor Diagnosis | [arXiv:2512.22184](https://arxiv.org/abs/2512.22184), Solo-authored, Dec 2025

- Motivated by whether AI-assisted diagnosis can be made reliable where high-resolution imaging infrastructure is unavailable. Built a 4-class brain MRI classification pipeline fusing **MobileNetV2** CNN embeddings with hand-crafted radiomics features (GLCM texture, shape, intensity) via late-fusion **Random Forest**; fusion achieved **95.1% validation accuracy** vs. 85.0% CNN-only baseline.
- Designed a robustness evaluation under controlled resolution degradation and synthetic Gaussian noise to simulate the image quality typical of low-resource clinical environments, testing not just accuracy on clean data but graceful degradation under realistic conditions. Explainability via **Grad-CAM** and feature importance analysis to surface which signals drove predictions.
- Evaluated against a CNN-only baseline to isolate the marginal contribution of the radiomics signal, motivated by the broader question of which model components are actually responsible for predictive performance rather than assuming end-to-end accuracy tells the full story.

EXPERIENCE

Open Source Contributor, Applied AI Capstone (AI301)

Remote

CodePath × Anthropic | Capstone Tier | Claude Code

Jun 2026 – Present

- Contributed two merged PRs to production open-source repositories as part of CodePath's most advanced AI engineering tier, developed with **Anthropic**. Fixed a live defect in `graphql-hive/console` (PR #8166, merged): reproduced CI failures, traced root cause across TypeScript modules, extracted a pure testable helper, wrote 6 unit tests, and resolved a Prettier CI failure after maintainer feedback. Also added Claude Sonnet 5 support across `traceroot-ai/traceroot` (PR #1400, merged), an AI agent observability platform, covering system config, pricing metadata, and tests.

Emerging Engineer Empowerment (E³) Scholar

Remote

CodePath | AI Engineering, Full-Stack Development, Interview Prep

Aug 2024 – Present

- Four cohorts of structured engineering education over two years; Anthropic-supported coursework in prompt engineering, RAG pipelines, agentic workflows, and evaluation methodology, building both implementation skills and a systematic approach to validating AI system behavior.

PROJECTS

RAG Pipeline Exploration Suite | *Python, LangChain, LangGraph, OpenAI Embeddings, FAISS* | [GitHub](#)

- Built five standalone RAG systems (PDF Q&A, multi-document chat, web-grounded QA, tool-calling agent, LangGraph routing agent), varying one architectural parameter at a time to build systematic intuition about how chunking strategy, embedding model, and retrieval method each affect output quality, a methodology directly transferable to evaluating clinical AI system behavior.

Intueri, LLM-to-Code STEM Teaching Tool | *Python, FastAPI, Docker, GCP* | [AI ATL 2025](#)

- Built a containerized backend (Docker, Redis, MongoDB, GCP) converting plain-English prompts into rendered video explanations. Identified a systematic failure mode where generated outputs appeared correct but were not, and added validation to catch the failure class before reaching users, the same caution required when AI outputs inform clinical decisions.

Augmented Voice, Real-Time ASL Recognition | *Python, OpenCV, MediaPipe, TensorFlow* | [Emory Hacks 2025](#)

- Built a real-time accessibility prototype classifying ASL gestures and converting them to spoken audio. Diagnosed a live-hardware latency failure by isolating and timing each pipeline stage, identified inference as the bottleneck, and restored real-time performance before the judging deadline, the same diagnostic discipline required when deploying models under real-world resource constraints.

ACCOMPLISHMENTS

SCUDEM Outstanding Award | *SIMIODE International Competition, Fall 2023*

- Formed one of GSU's first SCUDEM teams; modeled projectile dynamics under drag and reaction-time constraints using differential equations; earned the highest distinction from an international judge panel.

Grand Integrator | *1st Place, GSU Integral Bee*

- Won the university-wide undergraduate Integral Bee among all GSU undergraduates.

President's List ×6 | *Georgia State University, Fall '22, '23, Summer '24, Fall '24, '25, Spring '26*

- 4.0 GPA in six semesters across introductory through upper-division CS and mathematics coursework.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++, Bash

AI / ML: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, OpenAI API, FAISS, OpenCV, MediaPipe, Transformers

Medical / Scientific Computing: radiomics feature extraction, Grad-CAM explainability, robustness evaluation under data degradation

Backend / Full Stack: FastAPI, Node.js, React, REST APIs, MongoDB, Redis, Docker

Cloud: AWS (Certified Cloud Practitioner), GCP, Git/GitHub, Linux